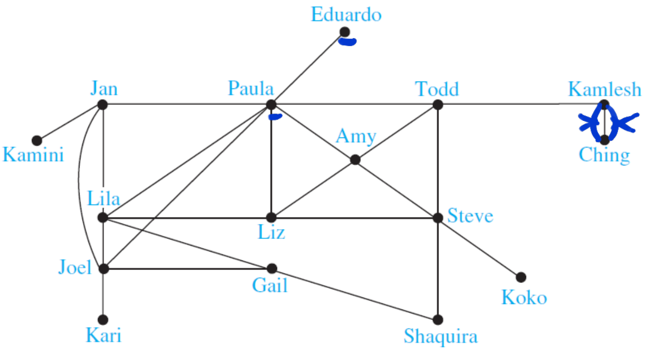
Graphs and Graph models

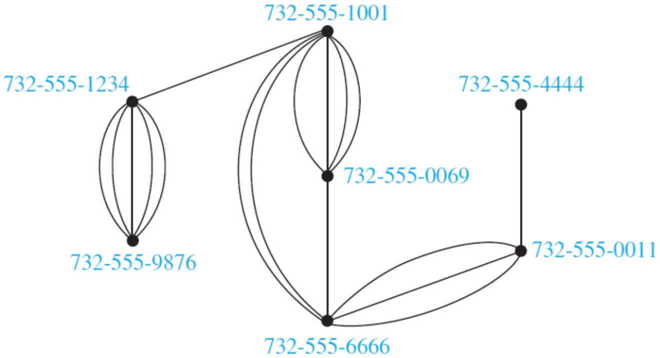
(Paragraph 10.1)

A graph $G = (V, E)$ consists of V , a nonempty set of vertices (or nodes) and E, a set of edges.
Each edge has either one or two vertices associated with it, called its endpoints.
An edge is said to connect its endpoints.



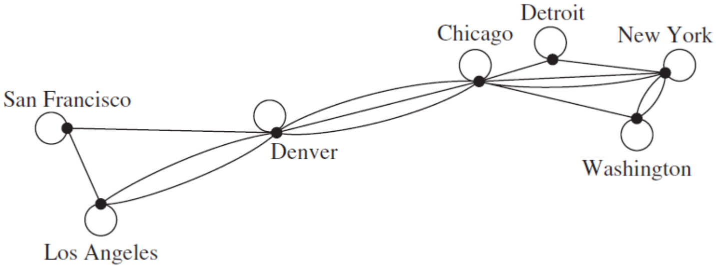
A graph in which each edge connects two different vertices and where no two edges connect the same pair of vertices is called a simple graph.

Graphs that may have multiple edges connecting the same vertices are called multigraphs.



Here graph models telephone calls. When we care only whether there has been a call connecting two telephone numbers, An edge connects telephone numbers when there has been a call between these numbers.

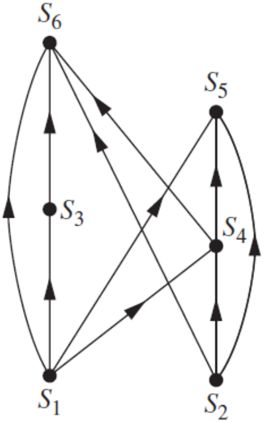
Edges that connect a vertex to itself are called loops, and sometimes we may even have more than one loop at a vertex.
Graphs that may include loops, and possibly multiple edges connecting the same pair of vertices or a vertex to itself, are called pseudographs.



Sometimes a communications link connects a data center with itself, perhaps a feedback loop for diagnostic purposes.

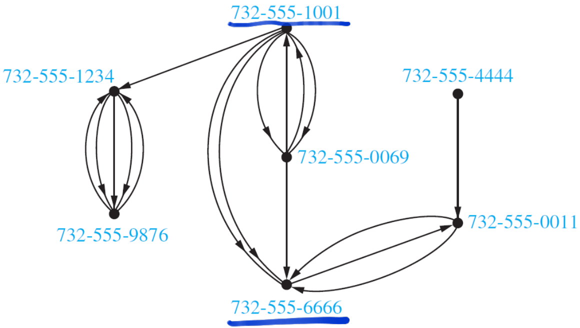
In a directed graph (or digraph) each directed edge is associated with an ordered pair of vertices.
The directed edge associated with the ordered pair (u, v) is said to start at u and end at v.

S_1	$a := 0$
S_2	$b := 1$
S_3	$c := a + 1$
S_4	$d := b + a$
S_5	$e := d + 1$
S_6	$e := c + d$



A Precedence Graph

Directed graphs that may have multiple directed edges from a vertex to a second (possibly the same) vertex are called directed multigraphs.
When there are m directed edges, each associated to an ordered pair of vertices (u, v), we say that (u, v) is an edge of multiplicity m.



The edge representing a call starts at the telephone number from which the call was made and ends at the telephone number to which the call was made.
For example, (732-555-1001, 732-555-6666) has multiplicity 2.

Graph terminology

Type	*Edges*	*Multiple Edges Allowed?*	*Loops Allowed?*
Simple graph	Undirected	No	No
Multigraph	Undirected	Yes	No
Pseudograph	Undirected	Yes	Yes
Simple directed graph	Directed	No	No
Directed multigraph	Directed	Yes	Yes
Mixed graph	Directed and undirected	Yes	Yes

Homework:
- Solve # 1,3,5,7,9,19,21,27,29,33 at p. 649-651.

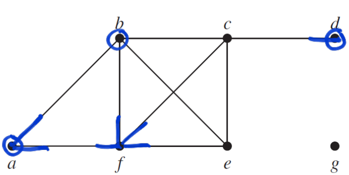
Answer Q1

Special types of graphs and graph isomorphism

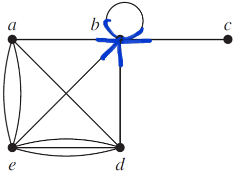
(Paragraphs 10.2 and 10.3)

Undirected graph

Two vertices u and v in an undirected graph G are called adjacent (or neighbors) in G if u and v are endpoints of an edge e of G.
Such an edge e is called incident with the vertices u and v.
The set of all neighbors v denoted by $N(v)$, is called the neighborhood of v.
The degree of a vertex is the number of edges incident with it, except that a loop at a vertex contributes twice to the degree of that vertex.
The degree of the vertex v is denoted by $\deg(v)$.



G



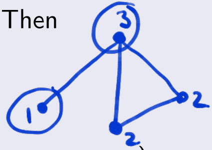
H

In G: $\deg(a) = 2$ $\deg(d) = 1$ $\deg(f) = 4$ $\deg(g) = 0$ (isolated vertex)
$N(a) = \{b, f\}$ $N(d) = \{c\}$ $N(f) = \{a, b, c, e\}$ $N(g) = \emptyset$
In H: $\deg(b) = 6$ $\deg(e) = 6$ $N(b) = \{a, b, c, d, e\}$ $N(e) = \{a, b, d\}$

THE HANDSHAKING THEOREM

Let $G = (V, E)$ be an undirected graph with m edges. Then

$$2m = \sum_{v \in V} \deg(v).$$



(Note that this applies even if multiple edges and loops are present.)

1, 2, 3, 4, 4, 5 - it is not possible

Theorem

An undirected graph has an even number of vertices of odd degree.

proof: V_1 - set of vertices with odd degrees

V_2 - set — " — even $\quad V = V_1 \cup V_2$

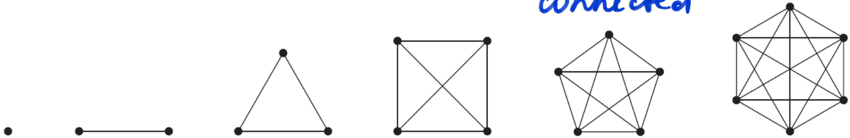
$$2m = \sum_{v \in V} \deg(v) = \sum_{v \in V_1} \deg(v) + \sum_{v \in V_2} \underbrace{\deg(v)}_{\text{even}} \leftarrow \text{even number}$$

$$\sum_{v \in V_1} \deg(v) = 2m - \text{even num} = \text{even number}$$

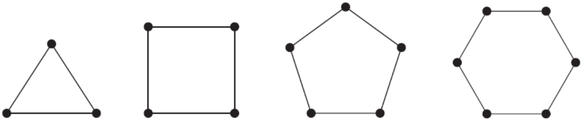
$\Rightarrow V_1$ contains even number of vertices $\blacksquare$

Some special simple graphs

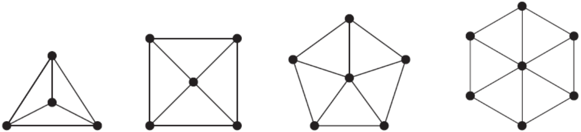
Complete graphs - all vertices are connected



K_1 K_2 K_3 K_4 K_5 K_6



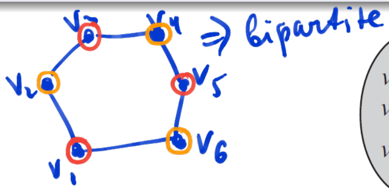
Cycles C_3 C_4 C_5 C_6

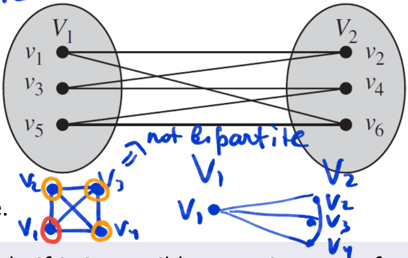


Wheels W_3 W_4 W_5 W_6

Bipartite graphs

A simple graph G is called bipartite if its vertex set V can be partitioned into two disjoint sets V_1 and V_2 such that every edge in the graph connects a vertex in V_1 and a vertex in V_2.





For example C_6 is bipartite.
Any complete graph is not bipartite.

A simple graph is bipartite if and only if it is possible to assign one of two different colors to each vertex of the graph so that no two adjacent vertices are assigned the same color.

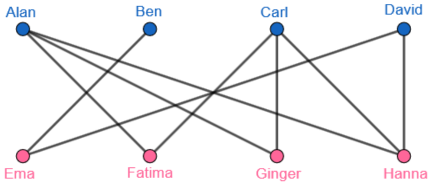
Answer Q2

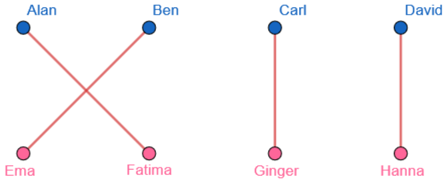
Graph matching problems are very common in daily activities: from online matchmaking and dating sites, to medical residency placement programs.

Suppose that there are n men and n women on an island.

Each person has a list of members of the opposite gender acceptable as a spouse.

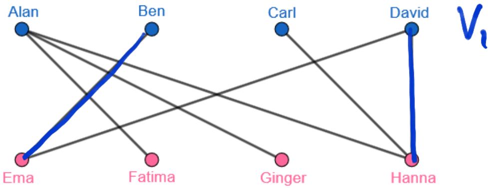
We construct a bipartite graph $G = (V_1, V_2)$ where V_1 is the set of men and V_2 is the set of women so that there is an edge between a man and a woman if they find each other acceptable as a spouse.





A complete matching is a set of married couples where every man is married.

Complete matching is not always possible



HALL'S MARRIAGE THEOREM (Philip Hall in 1935)

The bipartite graph $G = (V, E)$ with bipartition (V_1, V_2) has a complete matching from V_1 to V_2 if and only if $|\underline{N(A)}| \geq |A|$ for all subsets A of V_1.

In the example above take $A = \{\text{Ben, Carl, David}\}$.

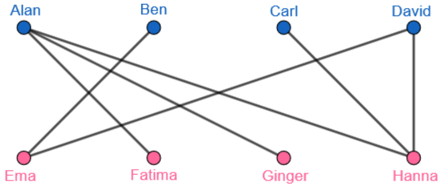
$N(A) = \{\text{Ema, Hanna}\}$ $\quad |N(A)| = 2, \quad |A| = 3$

$\Rightarrow$ the complete matching is not possible. $\quad 2 \not\geq 3$

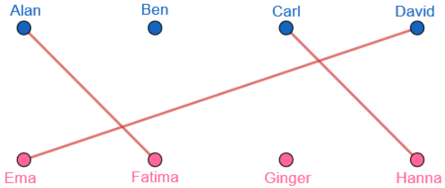
Maximal matching

A maximum matching is a largest possible set of married couples.

A maximal matching for



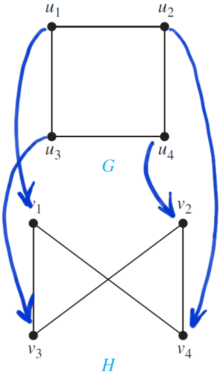
is



Maximal matching algorithms

- Hungarian Maximum Matching Algorithm or Kuhn-Munkres algorithm (1957, $O(n^3)$)
- Blossom Algorithm (1961, $O(|E|n^2)$, where $|E|$ is a number of edges)
- Hopcroft-Karp Algorithm (1973, $O(|E|\sqrt{n})$)

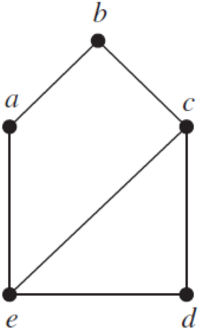
The simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are isomorphic if there exists a one-to- one and onto (bijection) function f from V_1 to V_2 with the property that a and b are adjacent in G_1 if and only if $f(a)$ and $f(b)$ are adjacent in G_2, for all a and b in V_1.
Such a function f is called an isomorphism.
Two simple graphs that are not isomorphic are called non-isomorphic.



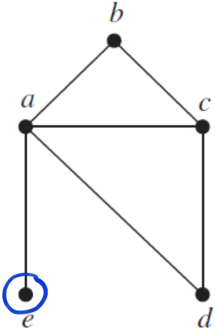
Two graphs G and H are isomorphic.
The isomorphism $f : G \to H$ is f is bijection
$f(u_1) = v_1$ $f(u_2) = v_4$ $f(u_3) = v_3$ and $f(u_4) = v_2$
check for adjacent edges ⇒ isomorph.
Is there other isomorphisms?
yes
For example, $f(u_1) = v_4$ $f(u_2) = v_2$ $f(u_3) = v_1$ and $f(u_4) = v_3$
It is enough to construct just one isom.

Show that two graphs below are not isomorphic.

1. $|V_1| = |V_2|$
$|E_1| = |E_2|$
2. The same degrees of vertices



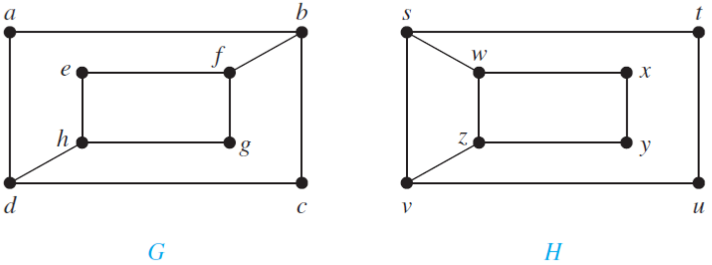
G



H

Solution: |The set of vertices| = 5 for both
|the set of edges| = 6 for both
H has $\deg(e) = 1$, G does not have $\Rightarrow$ not isomorphic.

Are the following graphs isomorphic?



Solution: 1. $|V_1| = |V_2| = 8$, $|E_1| = |E_2| = 10$

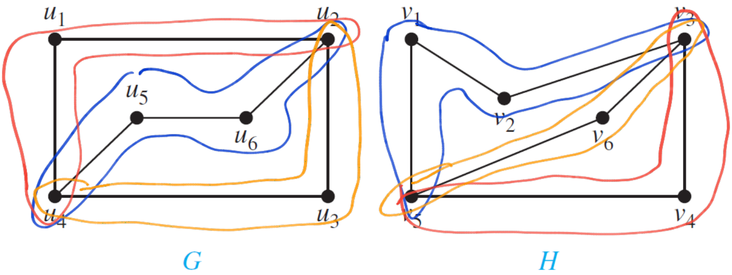
2. 4 vertices of deg 2, 4 vert. of deg 3
the same for G and H.

3. We must have the same chains

in G. 2 − 3 − 3   in H there is no chain 3-2-3

$\Rightarrow$ not isomorphic.

Are the following graphs isomorphic?



Solution: 1. $|V_1| = |V_2| = 6$, $|E_1| = |E_2| = 7$ ✓

2. 4 vert of 2 and 2 vert of 3 ✓ $\Rightarrow$ can try const. isomorph

3. $2 <^{3}_{3}$ — 2 chains 3-2-2-3 ✓

$f(u_1) = v_4$, $f(u_2) = v_3$, $f(u_3) = v_6$, $f(u_4) = v_5$,
$f(u_5) = v_1$, $f(u_6) = v_2$

Answer Q3

Algorithms for graph isomorphism

The best algorithms known for determining whether two graphs are isomorphic have exponential worst-case time complexity (in the number of vertices). However, linear average-case time complexity algorithms are known that solve this problem, and there is some hope, that an algorithm with polynomial worst-case time complexity for determining whether two graphs are isomorphic can be found.

Applications of graph isomorphism

- chemistry: use multigraphs, known as molecular graphs, to model chemical compounds. In these graphs, vertices represent atoms and edges represent chemical bonds between these atoms.
  When a potentially new chemical compound is synthesized, a database of molecular graphs is checked to see whether the molecular graph of the compound is the same as one already known.
- design of electronic circuits: using graphs in which vertices represent components and edges represent connections between them.
  Modern integrated circuits, known as chips, are miniaturized electronic circuits, often with millions of transistors and connections between them.
  Graph isomorphism is the basis for the verification that a particular a circuit produced corresponds to the original schematic of the design.

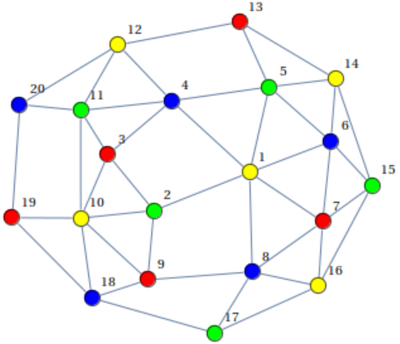
Homework:

- Solve exercises # 1,3,5,7,9,11,13,21,23,25,27,29 at p. 665-666.
- Read p. 668-671 of 10.3.
- Solve exercises # 1,3,5,7,9,11,13,15,17,19,21,23,35,37,3,41,43,45,55,57 at p. 675-677.

Graph Coloring

(Paragraph 10.8)

A coloring of a simple graph is the assignment of a color to each vertex of the graph so that no two adjacent vertices are assigned the same color.

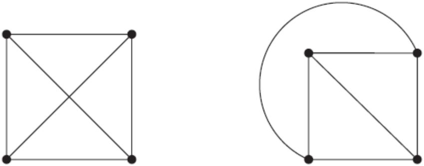


chi

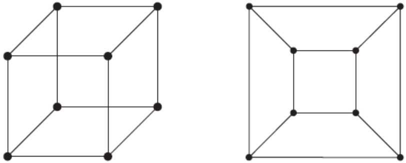
The chromatic number $\chi(G)$ of a graph G is the least number of colors needed for a coloring of this graph.

A graph is called planar if it can be drawn in the plane without any edges crossing.

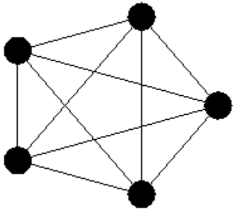
The graph K_4 without crossings:

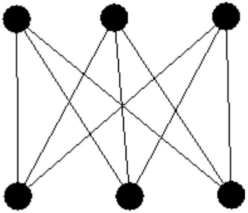


The graph Q_3 without crossings:



Examples of non-planar graph



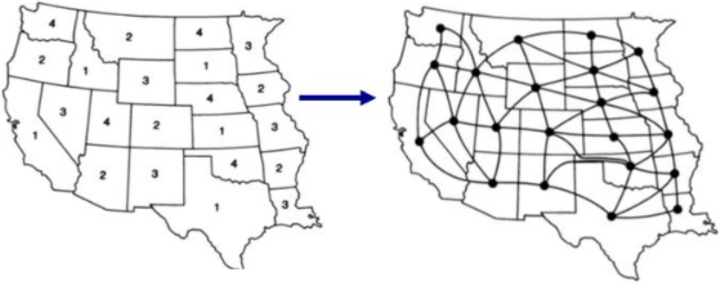


Kuratowski's Theorem

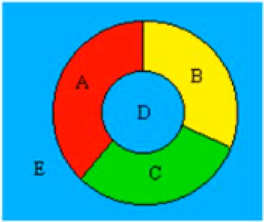
A graph is non-planar if and only if it contains a subgraph homeomorphic to $K_{3,3}$ or K_5.

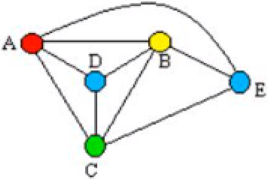
Maps vs graphs

If there is a map we can always draw a corresponding (planar) graph.
Only for planar graphs it is possible to construct a corresponding map.



Asking for the chromatic number of a planar graph is the same as asking for the minimum number of colors required to color a map so that no two adjacent regions are assigned the same color.





This question has been studied for more than 100 years.
The answer is provided by one of the most famous theorems in mathematics.

THE FOUR COLOR THEOREM

The chromatic number of a planar graph is no greater than four.

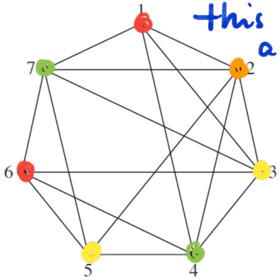
The four color theorem was originally posed as a conjecture in the 1850s. It was finally proved by the American mathematicians Kenneth Appel and Wolfgang Haken in 1976.

Scheduling Final Exam

How can the final exams at a university be scheduled so that no student has two exams at the same time?

Solution: This scheduling problem can be solved using a graph model, with vertices representing courses and with an edge between two vertices if there is a common student in the courses they represent.
Each time slot for a final exam is represented by a different color.
A scheduling of the exams corresponds to a coloring of the associated graph.

For instance, suppose there are seven finals to be scheduled. Suppose the courses are numbered 1 through 7. Suppose that the following pairs of courses have common students: 1 and 2, 1 and 3, 1 and 4, 1 and 7, 2 and 3, 2 and 4, 2 and 5, 2 and 7, 3 and 4, 3 and 6, 3 and 7, 4 and 5, 4 and 6, 5 and 6, 5 and 7, and 6 and 7.



this is a planar graph.

Because the chromatic number of this graph is 4, four time slots are needed.
The possible answer is I-1,6, II-2, III-3,5, IV-4,7

Homework:

- Solve exercises # 5,7,9,11,13,17,19 at p. 732-733.

Answer Q4